

Lesson 1: Learning from Data

45 minutes

Overview

In this lesson students explore the “What’s Going on in this Graph?” site in order to tell a “data story” which explains both what the data shows and why that might be. Following this, students are introduced to the concept of metadata and look for the metadata of datasets on App Lab.

Standards

Full Course Alignment

CSP Conceptual Framework

- **DAT-2** - Programs can be used to process data, which allows users to discover information and create new knowledge.

CSTA K-12 Computer Science Standards (2017)

- **DA** - Data & Analysis

Agenda

The Pre-Unit Pulse

Warm Up (5 minutes)

Activity (30 minutes)

What's Going On in This Graph?

Exploring Metadata

Wrap Up (10 minutes)

Objectives

Students will be able to:

- Differentiate between what data shows and why that might be the case
- Explain the usefulness of metadata

Preparation

- Preview **“What’s Going on in this Graph?”** and prepare for the demo
- Check the **“Teacher’s Lounge”** forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching this virtually, consider checking our **Virtual Modifications**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Unit 5 - Data** - Slides
▼ Make a Copy
- **CSP Unit 5 - Lesson 1 - Learning from Data** - Slides
▼ Make a Copy

For the students

- **Learning From Data** - Activity Guide
▼ Make a Copy
- **Unit 5 Journal (Digital)** - Resource
▼ Make a Copy
- **Unit 5 Journal (Interactive Notebook)** - Resource
▼ Make a Copy

- [What's Going On in This Graph?](#) - Resource

Vocabulary

- **Information** - the collection of facts and patterns extracted from data
- **Metadata** - data about data

Teaching Guide

The Pre-Unit Pulse

Have your students complete these questions independently some time before starting this unit.

☰ 1-3

The Pre-Unit Pulse

1

2

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💡 Teaching Tip

Pre-Unit Pulse Questions: The answers to these questions can provide insights into the preferences, strengths, and motivations of your students. If you'd like to adapt the questions or add in your own, you may want to make a copy of [this Google Form](#) which already has the existing questions populated.

For more tips and ideas of how to use these questions, check out the [CSP Guide to Pre-Unit Pulse Questions](#)

Warm Up (5 minutes)

Distribute: Give each student a copy of the Unit 5 Journal ([Digital/Interactive Notebook](#)) or have students set up this unit's section in their CSP journals.

💡 Teaching Tip

Journaling: Journaling can take many different forms, but in general, it's a tool for individual processing and reflection in a form that can be revisited as students develop their skills and understandings. The medium used for journaling can vary depending on classroom needs.

For more guidance and examples, check out the [CSP Guide to Journaling](#)

🎤 Remarks

Welcome to Unit 5: Data! In this unit, we are going to learn how to organize and visualize data to answer questions. We'll make charts, look for patterns, and consider the impact that data collection has on our world.

Let's get things started by asking some questions.

☰ **Discuss:** *What types of data visualizations do you know of? Why do we visualize data?*

Discussion Goal: Use this discussion to gauge some of the prior knowledge your students hold about data visualizations. Push students to think about how data visualizations help people answer questions, observe patterns, and generate new ideas and questions.

Activity (30 minutes)

What's Going On in This Graph? (20 minutes)

Remarks

Today you are going to get practice in telling a data story based on a data visualization. Let's start by looking at an example of a data visualization.

 **Display:** The graph showing Fast-food restaurants Population 16-19, and Teenage Employment.

 **Discuss:** *What do you notice? What do you wonder? What might be going on in this graph?*

Discussion Goal: Use this discussion to help point out features of the graph like the scale used on the y-axis and the fact that the percentage change is compared with the year 2000. Also, use this discussion to get students started thinking about what is shown in the graph and what outside knowledge they might be bringing in to try and understand what might be going on in this graph.

Remarks

This chart gives us information: a collection of facts and patterns extracted from data. We can use this information to identify trends, make connections and address problems.

 There are two distinctions we need to make when looking at a chart or visualization:

- What does the data show?
- Why might that be the case?

The "what" is the facts of the matter. The percentage of teenagers in the workforce has decreased since 2000. The "why" is an informed opinion. Fewer teenagers are working because more are focused on their education and earning scholarships.

Teaching Tip

One of the main goals of this lesson is getting students acquainted with talking and writing about data. In particular we want to:

- Draw a distinction between describing what the data shows and describing why it might be that way
- In other words: describe connections and trends in data separate from drawing conclusions.
- We want students to get in the habit of separating the what from the why when it comes to talking and writing about data

 **Demo:** Demonstrate how to navigate the "What's Going On in This Graph?" website in front of the class. Show your students the two ways the graphs are sorted (by topic and by type). Then, show them the page for the **"A Fast-Food Problem: Where Have All the Teenagers Gone?"** and walk them through the structure of the page and instruct them just focus on the questions for now. If time allows, you may choose to demo another graph and model stating the What and the Why as you make look at the charts with the class.

 **Do This:** In pairs, have students use the "What's Going On in This Graph?" site to look for "data stories" and have them complete the Activity Guide. Depending on class time, allow students to share their data stories with another group or with the class as a whole.

While the resource for this lesson contains 60 graphs, the "What's Going On in This Graph?" **column** from the New York Times is still updated and may be something you'd like to share with your students for more current graphs.

Remarks

As you were determining the **Why** in your data stories, you may have been tempted to draw concrete connections. However, it's important to remember that correlation (similarities, patterns) does not equal causation (this thing caused that thing). There can be any number of reasons why a pattern or interesting data point may appear in a chart - and our job is to make an informed decision while recognizing that there may be multiple factors at play. Usually additional research with several data sets is necessary to understand the exact nature of the relationship between data. Did this one thing cause another thing to happen? Do more people search for chocolate because they want to give it as a gift, or could there perhaps be another reason?

These two graphs show examples of visualizations that show a correlation between two factors, but it would be pretty hard to make the case that either of these things caused the other. These are some examples that are meant to be silly, but also they show us the importance of separating out the "What" from the "Why".

In this unit we will be making charts to help answer questions:

- *"I think this visualization tells me this..."*
 - Something is more popular than something else
 - Something is more important than something else
 - Something has become more or less searched over time
- *"... but I'm not sure because..."*
 - I don't know exactly how the data was collected
 - This might tell me people searched for green more than red, but it doesn't tell me why they do that or that green is a better color
 - We need more data!

Exploring Metadata (10 minutes)

Remarks

When we consider datasets, it's helpful to know as much about those datasets as possible. Where did the data come from? How much data is included? When was it collected?

All of this information is considered **metadata** which is defined as "data about data".

We can have metadata about any digital data. For example, this picture contains metadata that tells us the when the picture was created, what the resolution is, and how many people have downloaded it. Datasets can also have metadata that explains more about the information in the dataset. You've seen this in App Lab!

Discuss: *What is the metadata for the graph you picked for your data story?*

Discussion Goal: If students haven't already found it, point them to the bottom of the page where they found their graph where the source of the data and some other information about it can be found. If students want to dig deeper, most of the graphs have an article associated with them they could read for more context.

Remarks

Now let's explore metadata in App Lab datasets. Where can we find this information? What can it tell us about the data? How can it help us organize the data?

 **Do This:** Students navigate to Level 4 on Code Studio where they open the data tab and look at the metadata for a table. Then they should share with a partner where the data comes from and what they can learn from the metadata.



Exploring Metadata

Teaching Tip

You may also want to connect this concept of metadata to things that students use in their daily lives. For example, these websites show the steps to view the metadata on photos taken with [iPhones](#) and [Android](#) devices.

Wrap Up (10 minutes)

 **Review:** Review key takeaways on the slide. Students may want to jot down notes in their journals.

 **Journal:** Students add the following word and definition to their journal: metadata.

Assessment: Check For Understanding

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: Google Trends which is a tool that allows you to visualize data about search history across different times and locations.

Below is an image from Google Trends that plots Cats and Dogs. Choose the most accurate description of what this data is actually showing based on what you know about how Google Trends works.

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Check For Understanding



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